

Preliminary

Theorem 1.1 (Cauchy Schwarz). Suppose X and Y are random variables, then:

$$\mathbb{C}^2(X, Y) \leq \mathbb{V}(X) \cdot \mathbb{V}(Y), \quad (1.1)$$

where equality holds $\iff Y$ and X are linearly related, $Y = aX + b$. $\square$

1.1 Exercises

- i. Prove (1.1) starting from the observation that $0 \leq \mathbb{V}(tX + Y)$ for a constant t .

Data Reduction

2.1 Notation

Suppose $\mathbf{Y} = (Y_1, \dots, Y_n)$ is a random sample of size n , with realization $\mathbf{y} = (y_1, \dots, y_n)$, from a distribution with joint density $f(\mathbf{y}|\theta) = f(y_1, \dots, y_n|\theta)$.

2.2 Sufficiency

Definition 2.1. A statistic T is **sufficient** for θ if the conditional distribution of the sample $\mathbf{Y}$ given T does not depend on θ . $\blacksquare$

Theorem 2.1 (Factorization). A statistic $T(\mathbf{y})$ is sufficient for $\theta \iff$ for $\forall(\mathbf{y}, \theta)$ the joint density factors as:

$$f(\mathbf{y}|\theta) = g\{T(\mathbf{y})|\theta\}h(\mathbf{y}).$$

$\square$

Proof. ($\implies$) If T is sufficient for θ , then $\mathbb{P}\{\mathbf{Y} = \mathbf{y} | T = t(\mathbf{y})\}$ does not depend on θ . Express the joint density as:

$$\begin{aligned} f(\mathbf{y}|\theta) &= \mathbb{P}(\mathbf{Y} = \mathbf{y}|\theta) \\ &= \mathbb{P}\{\mathbf{Y} = \mathbf{y} \cap T = t(\mathbf{y})|\theta\} \\ &= \mathbb{P}\{T = t(\mathbf{y})|\theta\} \cdot \mathbb{P}\{\mathbf{Y} = \mathbf{y} | T = t(\mathbf{y})\} \\ &= g(T|\theta) \cdot h(\mathbf{y}). \end{aligned}$$

($\impliedby$) Suppose the factorization exists. Define the subset $\mathcal{A}(\mathbf{y})$ of the sample space $\mathcal{Y}$:

$$\mathcal{A}(\mathbf{y}) = \{\mathbf{u} \in \mathcal{Y} : t(\mathbf{u}) = t(\mathbf{y})\}.$$

That is, $\mathcal{A}(\mathbf{y})$ contains those realizations of $\mathbf{Y}$ that lead to the same sufficient statistic as $\mathbf{y}$. The density of T is expressible as:

$$\mathbb{P}\{T = t(\mathbf{y})|\theta\} = \sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} f(\mathbf{u}|\theta) = \sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} g\{t(\mathbf{y})|\theta\}h(\mathbf{u}) = g\{t(\mathbf{y})|\theta\} \sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} h(\mathbf{u}).$$

The distribution of the data given T is:

$$\begin{aligned} \mathbb{P}\{\mathbf{Y} = \mathbf{y}|T = t(\mathbf{y})\} &= \frac{\mathbb{P}\{\mathbf{Y} = \mathbf{y} \cap T = t(\mathbf{y})\}}{\mathbb{P}\{T = t(\mathbf{y})\}} \\ &\stackrel{*}{=} \frac{\mathbb{P}(\mathbf{Y} = \mathbf{y})}{\mathbb{P}\{T = t(\mathbf{y})\}} \\ &= \frac{g\{t(\mathbf{y})|\theta\}h(\mathbf{y})}{g\{t(\mathbf{y})|\theta\} \sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} h(\mathbf{u})} \\ &= \frac{h(\mathbf{y})}{\sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} h(\mathbf{u})}. \end{aligned}$$

Equality $\stackrel{*}{=}$ follows since the event $\{\mathbf{Y} = \mathbf{y}\}$ is a subset of the event $\{T = t(\mathbf{y})\}$. That is, $\mathbf{Y} = \mathbf{y} \implies T = t(\mathbf{y})$, but not conversely. ■

Definition 2.2. An **exponential family** density takes the form:

$$f(y|\theta) = h(y)c(\theta) \exp \left\{ \sum_{k=1}^K \omega_k(\theta) t_k(y) \right\}, \quad (2.1)$$

with the support of y not depending on θ .

The **canonical parameterization** of (2.1) is:

$$f(y|\eta) = h(y)c(\eta) \exp \left\{ \sum_{k=1}^K \eta_k t_k(y) \right\}.$$

If the parameter space of η includes an open K -dimensional rectangle, then the exponential family is **full-rank**. Otherwise, it is **curved**. ■

Theorem 2.2 (Exponential Family). Suppose each Y_i follows an exponential family distribution (2.1), then the sufficient statistics for θ are:

$$\mathbf{T} = \left(\sum_{i=1}^n t_1(Y_i), \dots, \sum_{i=1}^n t_K(Y_i) \right).$$

If the exponential family has full rank, then $\mathbf{T}$ is also complete. See Casella & Berger (2002) 6.2.10 and 6.2.25. □

2.3 Completeness

Definition 2.3. A statistic T is **complete** if $\mathbb{E}\{g(T)\} = 0$ for $\forall \theta \implies g(T) = 0$ with probability one. ■

Definition 2.4. A statistic A whose distribution does not depend on θ is **ancillary**. ■

Theorem 2.3 (Basu's). If T is a complete sufficient statistic, then T is independent of every ancillary statistic. □

Proof. Suppose A is ancillary for θ , and that T is complete and sufficient. Since A is ancillary, $\mathbb{P}(A = a)$ does not depend on θ . Define the subset $\mathcal{A}(\mathbf{y})$ of $\mathcal{Y}$:

$$\mathcal{A}(\mathbf{y}) = \{\mathbf{u} \in \mathcal{Y} : a(\mathbf{u}) = a(\mathbf{y})\}.$$

The distribution of A given T is expressible as:

$$\mathbb{P}\{A = a(\mathbf{y}) | T = t(\mathbf{y})\} = \sum_{\mathbf{u} \in \mathcal{A}(\mathbf{y})} \mathbb{P}\{\mathbf{Y} = \mathbf{u} | T = t(\mathbf{y})\}.$$

Since T is *sufficient*, $\mathbb{P}\{\mathbf{Y} = \mathbf{u} | T = t(\mathbf{y})\}$ does not depend on θ , therefore neither does $\mathbb{P}\{A = a(\mathbf{y}) | T = t(\mathbf{y})\}$. Define:

$$g(t) = \mathbb{P}(A = a | T = t) - \mathbb{P}(A = a).$$

Since neither $\mathbb{P}(A = a | T = t)$ (by sufficiency) or $\mathbb{P}(A = a)$ (by ancillarity) depend on θ , $g(T)$ is a valid statistic. By iterated expectation:

$$\mathbb{E}\{g(T)\} = \mathbb{E}\{\mathbb{P}(A = a | T = t)\} - \mathbb{P}(A = a) = \mathbb{P}(A = a) - \mathbb{P}(A = a) = 0.$$

Since T is complete, $\mathbb{P}(A = a | T = t) = \mathbb{P}(A = a)$ with probability one. Conclude that A is independent of T . ■

2.4 Exercises

- i. Suppose $Y_i \sim N(\mu, \sigma^2)$. Show that $(\bar{Y}, S^2)$ are sufficient for (μ, σ^2) .
- ii. Suppose $Y_i \sim U(0, \theta)$. Show that $\max_i Y_i$ is complete and sufficient for θ .
- iii. Suppose $Y_i \sim g(y - \theta)$. Show that $Y_{(n)} - Y_{(1)}$ is ancillary for θ .
- iv. Suppose $Y_i \sim \theta^{-1}g(\theta^{-1}y)$. Show that $Y_i/\bar{Y}$ is ancillary for θ .
- v. Find the complete and sufficient statistics for these distributions:

- (a) Binomial.

(b) Poisson.

(c) Gamma.

vi. (**Exponential family**):

(a) Show that for a canonical-form exponential family distribution:

$$c(\eta) = \left(\int h(\mathbf{y}) \exp \left\{ \sum_{k=1}^K \eta_k t_k(y) \right\} dy \right)^{-1}.$$

(b) Derive the moment generating function of the canonical-form exponential family distribution.

(c) Obtain expressions for $\mathbb{E}\{t_k(Y)\}$ and $\mathbb{C}\{t_k(Y), t_l(Y)\}$, $k \neq l$.

Estimation

Definition 3.1. An **estimator** is a statistic, a random function of the data, intended to estimate a parameter θ . An **estimate** is a realization of an estimator. ■

Discussion 3.1 (Satterthwaite Approximation). **Method of moments** is a technique for deriving estimators in which sample moments are matched with population moments to obtain a system of simultaneous equations. Suppose $Y_i \sim \chi_{\nu_i}^2(0)$. Consider approximating the distribution of $T = \sum_{i=1}^n \omega_i Y_i$, where the ω_i are known weights, by a $\chi_\nu^2(0)$ distribution. In particular, the problem is to find ν such that:

$$T = \sum_{i=1}^n \omega_i Y_i \sim \frac{\chi_\nu^2(0)}{\nu}.$$

Equating $\mathbb{E}(T) = \sum_{i=1}^n \omega_i \nu_i$ with $\mathbb{E}(\chi_\nu^2/\nu) = 1$ gives the constraint:

$$\sum_{i=1}^n \omega_i \nu_i = 1. \quad (3.1)$$

The second moment of the $\chi_\nu^2(0)$ distribution is $\mathbb{E}\{(\chi_\nu^2)^2\} = \nu(\nu + 2)$. Equating $\mathbb{E}(T^2)$ with $\mathbb{E}\{(\chi_\nu^2)^2/\nu^2\} = 1 + 2/\nu$ and solving for ν gives:

$$\hat{\nu} = \frac{2}{\hat{\mathbb{E}}(T^2) - 1} = \frac{2}{(\sum_{i=1}^n \omega_i Y_i)^2 - 1}. \quad (3.2)$$

Since the estimator in (3.2) can be negative, consider instead:

$$\begin{aligned} \mathbb{E}(T^2) &= \mathbb{V}(T) + \mathbb{E}^2(T) \\ &= \mathbb{E}^2(T) \left\{ \frac{\mathbb{V}(T)}{\mathbb{E}^2(T)} + 1 \right\}. \end{aligned}$$

Setting the leading factor of $\mathbb{E}^2(T) \stackrel{\text{Set}}{=} 1$, since $\mathbb{E}(T) = 1$ under (3.1), and equating:

$$\mathbb{E} \left\{ \frac{(\chi_\nu^2)^2}{\nu^2} \right\} = 1 + \frac{2}{\nu} \stackrel{\text{Set}}{=} \left\{ \frac{\mathbb{V}(T)}{\mathbb{E}^2(T)} + 1 \right\},$$

gives the improved estimator:

$$\hat{\nu} = \frac{2\hat{\mathbb{E}}^2(T)}{\hat{\mathbb{V}}(T)}. \quad (3.3)$$

The numerator may be approximated:

$$\hat{\mathbb{E}}(T) = \sum_{i=1}^n \omega_i Y_i.$$

Taking the variance of T analytically:

$$\mathbb{V} \left(\sum_{i=1}^n \omega_i Y_i \right) \stackrel{\text{IND}}{=} \sum_{i=1}^n \omega_i^2 \mathbb{V}(Y_i) = \sum_{i=1}^n \omega_i^2 \cdot 2\nu_i \stackrel{*}{=} 2 \sum_{i=1}^n \omega_i^2 \cdot \frac{\mathbb{E}^2(Y_i)}{\nu_i},$$

where equality $\stackrel{*}{=}$ follows from $\mathbb{E}(Y_i) = \nu_i$. Making the approximation:

$$\hat{\mathbb{V}}(T) = 2 \sum_{i=1}^n \omega_i^2 \cdot \frac{\hat{\mathbb{E}}^2(Y_i)}{\nu_i} = 2 \sum_{i=1}^n \omega_i^2 \frac{Y_i^2}{\nu_i},$$

the final form of the Satterthwaite estimator in (3.3) is:

$$\hat{\nu} = \frac{(\sum_{i=1}^n \omega_i Y_i)^2}{\sum_{i=1}^n \omega_i^2 \frac{Y_i^2}{\nu_i}}. \quad (3.4)$$

(Source: Casella & Berger, 7.2.3.)



3.1 Likelihood

Definition 3.2. The **likelihood** $L(\theta|\mathbf{y}) = f(\mathbf{y}|\theta)$ is the joint density of the observed data viewed as a function of θ . The *log likelihood* is denoted:

$$\ell_n(\theta) \equiv \ln f(\mathbf{y}|\theta).$$

The **maximum likelihood estimate** (MLE) of θ maximizes the log likelihood:

$$\hat{\theta}_n \equiv \arg \max_{\theta \in \Theta} \ell_n(\theta).$$

The sample **score** for θ is the gradient of the log likelihood with respect to θ :

$$\mathcal{U}_n(\theta) \equiv \frac{\partial \ell_n}{\partial \theta}.$$

Here the subscript n distinguishes the sample score from the unit score:

$$u_i(\theta) \equiv \frac{\partial}{\partial \theta} \ln f(y_i|\theta).$$

The MLE is often obtained by solving the score equations:

$$\mathcal{U}_n(\theta) \stackrel{\text{Set}}{=} 0.$$

The *Hessian* for θ is the second derivative of the log likelihood in θ :

$$\mathcal{H}_n(\theta) \equiv \frac{\partial^2 \ell_n}{\partial \theta \partial \theta'}$$

The **observed information** for θ is the negative Hessian:

$$\mathcal{J}_n(\theta) \equiv -\mathcal{H}_n(\theta).$$

The **Fisher information** is the variance of the score:

$$\mathcal{I}_n(\theta) \equiv \mathbb{V}\{\mathcal{U}_n(\theta)\}.$$

The unit Fisher information is the variance of the unit score:

$$\iota(\theta) \equiv \mathbb{V}\{u_i(\theta)\}.$$

For exponential family distributions, the Fisher information coincides with the negative expected Hessian:

$$\mathcal{I}_n(\theta) \stackrel{*}{=} -\mathbb{E}\{\mathcal{H}_n(\theta)\}.$$

■

Theorem 3.1 (Asymptotic Normality). For $Y_i \stackrel{\text{iid}}{\sim} f(y|\theta)$, suppose the following conditions are satisfied:

- θ is an interior point of the parameter space Θ .
- θ is *identified*, meaning $\theta_1 \neq \theta_2$ implies $F(y|\theta_1) \neq F(y|\theta_2)$ for at least some y .
- The first 3 partial derivatives of $\ell(\theta)$ exist for y in the support of $F(y|\theta)$.
- The 3rd derivative of $\ell(\theta)$ is dominated element-wise by an integrable $g(y)$:

$$\left| \frac{\partial^3 \ln f(y|\theta)}{\partial \theta_i \partial \theta_j \partial \theta_k} \right| \leq g(y),$$

where $\int g(y) dF(y|\theta_0) < \infty$.

- For $\theta \in \Theta$, the unit score has expectation zero $\mathbb{E}\{u_i(\theta)\} = 0$, and the unit Fisher information $\iota(\theta) = \mathbb{V}\{u_i(\theta)\}$ is positive definite.
- The solution $\hat{\theta}_n$ to the sample score equation $\mathcal{U}_n(\theta) \stackrel{\text{Set}}{=} 0$ is consistent for θ , meaning:

$$\lim_{n \rightarrow \infty} \mathbb{P}\{||\hat{\theta}_n - \theta|| > \epsilon\} = 0.$$

Then for $n \rightarrow \infty$:

$$\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow N(0, \iota^{-1}), \quad (3.5)$$

where the limiting variance is the *inverse unit Fisher information*. $\square$

Proof. The proof follows from asymptotic normality of M-estimators. See (e.g.) Boos and Stefanski (2013) theorem 7.2. $\blacksquare$

Lemma 3.1 (Invariance Principle). If $\hat{\theta}_n$ maximizes the log likelihood $\ell_n(\theta)$ and $\tau(\theta)$ is some function of θ , then the MLE of τ is $\hat{\tau}_n = \tau(\hat{\theta}_n)$. $\blacksquare$

Proof. Since τ is not necessarily bijective, the *induced log likelihood* of τ is *defined* as:

$$\ell_n^*(t) = \sup_{\{\theta: \tau(\theta)=t\}} \ell_n(\theta).$$

Since the iterated maximization is equal to unconditional maximization:

$$\sup_{t \in \mathcal{T}} \ell_n^*(t) = \sup_{t \in \mathcal{T}} \sup_{\{\theta: \tau(\theta)=t\}} \ell_n(\theta) = \sup_{\theta \in \Theta} \ell_n(\theta).$$

That is, the maximum of the induced log likelihood coincides with the maximum of the original log likelihood: $\ell_n^*(\hat{\tau}_n) = \ell_n(\hat{\theta}_n)$. Finally, since $\ell_n(\hat{\theta}_n)$ is expressible as:

$$\ell_n(\hat{\theta}_n) = \sup_{\{\theta: \tau(\theta)=\tau(\hat{\theta}_n)\}} \ell_n(\theta),$$

and by definition:

$$\sup_{\{\theta: \tau(\theta)=\tau(\hat{\theta}_n)\}} \ell_n(\theta) = \ell_n^*\{\tau(\hat{\theta}_n)\},$$

conclude that $\ell_n^*(\hat{\tau}_n) = \ell_n^*\{\tau(\hat{\theta}_n)\}$, or $\hat{\tau}_n = \tau(\hat{\theta}_n)$. $\blacksquare$

3.2 Evaluation of Estimators

Definition 3.3. The **mean squared error** (MSE) of an estimator $\hat{\theta}$ of θ is:

$$\text{MSE} = \mathbb{E}(\hat{\theta} - \theta)^2.$$

$\blacksquare$

Definition 3.4. The bias of an estimator is the difference between its expectation and the true parameter:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta.$$

■

Lemma 3.2 (Bias-Variance Decomposition). The MSE of an estimator decomposes as:

$$\text{MSE} = \mathbb{V}(\hat{\theta}) + \text{Bias}^2(\hat{\theta}). \quad (3.6)$$

In the case of an *unbiased* estimator, the MSE is the variance. ■

Definition 3.5. $\hat{\theta}$ is the **uniform minimum variance unbiased estimator** (UMVUE) of θ if $\mathbb{E}(\hat{\theta}) = \theta$, and for any other estimator $\tilde{\theta}$ with $\mathbb{E}(\tilde{\theta}) = \theta$:

$$\mathbb{V}(\hat{\theta}) \leq \mathbb{V}(\tilde{\theta}).$$

■

Theorem 3.2 (Uniqueness). If the UMVUE of θ exists, then it is unique. □

Proof. Suppose not. Let $\hat{\theta}_1$ and $\hat{\theta}_2$ denote two UMVUEs of θ . Define:

$$\bar{\theta} = \frac{\hat{\theta}_1}{2} + \frac{\hat{\theta}_2}{2}.$$

Let $\varsigma^2 = \mathbb{V}(\hat{\theta}_1) = \mathbb{V}(\hat{\theta}_2)$. The variance of $\bar{\theta}$:

$$\begin{aligned} \mathbb{V}(\bar{\theta}) &= \frac{1}{4}\mathbb{V}(\hat{\theta}_1) + \frac{1}{4}\mathbb{V}(\hat{\theta}_2) + \frac{1}{2}\mathbb{C}(\hat{\theta}_1, \hat{\theta}_2) \\ &\stackrel{*}{\leq} \frac{1}{4}\varsigma^2 + \frac{1}{4}\varsigma^2 + \frac{1}{2}\sqrt{\mathbb{V}(\hat{\theta}_1)\mathbb{V}(\hat{\theta}_2)} = \varsigma^2, \end{aligned}$$

where $\stackrel{*}{\leq}$ is an application of the Cauchy-Schwarz inequality (1.1). If the inequality is strict, then neither $\hat{\theta}_1$ nor $\hat{\theta}_2$ is a UMVUE. Otherwise, $\hat{\theta}_2 = a\hat{\theta}_1 + b$, but then:

$$\mathbb{C}(\hat{\theta}_1, \hat{\theta}_2) = a\mathbb{V}(\hat{\theta}_1) = a\varsigma^2 \implies a = 1.$$

Moreover, to maintain unbiasedness:

$$\mathbb{E}(\hat{\theta}_2) = \mathbb{E}(\hat{\theta}_1) + b = \theta + b \implies b = 0.$$

Conclude that $\hat{\theta}_2 = \hat{\theta}_1$. ■

3.3 Cramer Rao Lower Bound

Theorem 3.3 (Cramer Rao Lower Bound). Suppose that $\mathbf{Y}$ is a random sample of size n , and that $\hat{\theta} = \hat{\theta}(\mathbf{Y})$ is an estimator satisfying:

$$\frac{d}{d\theta} \mathbb{E}(\hat{\theta}) = \int \frac{\partial}{\partial \theta} \{ \hat{\theta}(\mathbf{y}) f(\mathbf{y}|\theta) \} d\mathbf{y}, \quad (3.7)$$

and $\mathbb{V}(\hat{\theta}) < \infty$. Then:

$$\mathbb{V}(\hat{\theta}) \geq \frac{\left\{ \frac{d}{d\theta} \mathbb{E}(\hat{\theta}) \right\}^2}{\mathbb{E} \left\{ \frac{d}{d\theta} \ln f(\mathbf{y}|\theta) \right\}^2}. \quad (3.8)$$

□

Proof. Applying (3.7):

$$\begin{aligned} \frac{d}{d\theta} \mathbb{E}(\hat{\theta}) &= \int \frac{\partial}{\partial \theta} \{ \hat{\theta}(\mathbf{y}) f(\mathbf{y}|\theta) \} d\mathbf{y} = \int \hat{\theta}(\mathbf{y}) \frac{\partial f(\mathbf{y}|\theta)}{\partial \theta} d\mathbf{y} \\ &= \int \hat{\theta}(\mathbf{y}) \frac{\frac{\partial f(\mathbf{y}|\theta)}{\partial \theta}}{f(\mathbf{y}|\theta)} \cdot f(\mathbf{y}|\theta) d\mathbf{y} \\ &= \mathbb{E} \left\{ \hat{\theta}(\mathbf{y}) \cdot \frac{\partial}{\partial \theta} \ln f(\mathbf{y}|\theta) \right\}. \end{aligned}$$

Identify $\partial_\theta \ln f(\mathbf{y}|\theta)$ as the sample score for θ :

$$\frac{d}{d\theta} \mathbb{E}(\hat{\theta}) = \mathbb{E} \left\{ \hat{\theta}(\mathbf{y}) \cdot \mathcal{U}_n(\theta) \right\}.$$

Since the score has expectation zero ($\mathbb{E}\{\mathcal{U}_n(\theta)\} = 0$):

$$\frac{d}{d\theta} \mathbb{E}(\hat{\theta}) = \mathbb{E} \left\{ \hat{\theta}(\mathbf{y}) \cdot \mathcal{U}_n(\theta) \right\} = \mathbb{C} \{ \hat{\theta}(\mathbf{y}), \mathcal{U}_n(\theta) \}.$$

Likewise, since the score has expectation zero:

$$\mathbb{V} \{ \mathcal{U}_n(\theta) \} = \mathbb{E} \{ \mathcal{U}_n^2(\theta) \}.$$

Applying the Cauchy-Schwarz inequality (1.1) to $X = \hat{\theta}$ and $Y = \mathcal{U}_n(\theta)$:

$$\left\{ \frac{d}{d\theta} \mathbb{E}(\hat{\theta}) \right\}^2 = \mathbb{C}^2 \{ \hat{\theta}(\mathbf{y}), \mathcal{U}_n(\theta) \} \leq \mathbb{V}(\hat{\theta}) \cdot \mathbb{V} \{ \mathcal{U}_n(\theta) \} = \mathbb{V}(\hat{\theta}) \cdot \mathbb{E} \{ \mathcal{U}_n^2(\theta) \}.$$

■

Remark 3.1. The denominator of the Cramer Rao lower bound (CRLB) (3.8) is the variance of the score, which is the Fisher information:

$$\mathbb{E} \left\{ \frac{d}{d\theta} \ln f(\mathbf{y}|\theta) \right\}^2 = \mathbb{E} \{ \mathcal{U}_n^2(\theta) \} = \mathbb{V} \{ \mathcal{U}_n^2(\theta) \} = \mathcal{I}_n(\theta).$$

In the case of an unbiased estimator $\mathbb{E}\{\hat{\theta}\} = \theta$, the CRLB reduces to:

$$\mathbb{V}(\hat{\theta}) \geq \mathcal{I}_n^{-1}(\theta).$$

If the sample is IID, then $\mathcal{I}_n(\theta) = n\mathcal{I}(\theta)$, and:

$$\mathbb{V}(\hat{\theta}) \geq \{n\mathcal{I}(\theta)\}^{-1} \quad (3.9)$$

The right hand side of (3.9) is identically the limiting variance of the MLE (3.5).

The CRLB only applies when differentiation in θ commutes with integration in $\mathbf{y}$ (3.7). In general, this condition will fail when the support of $\mathbf{y}$ depends on θ . $\blacklozenge$

Lemma 3.3 (Fisher Information). If $\mathbf{Y}$ is a random sample from a density $f(\mathbf{y}|\theta)$ that satisfies:

$$\frac{d}{d\theta}\mathbb{E}\{\mathcal{U}_n(\theta)\} = \frac{d}{d\theta}\mathbb{E}\left\{\frac{\partial}{\partial\theta}\ln f(\mathbf{y}|\theta)\right\} = \int \frac{\partial}{\partial\theta} \left[\left\{\frac{\partial}{\partial\theta}\ln f(\mathbf{y}|\theta)\right\} f(\mathbf{y}|\theta) \right] d\mathbf{y},$$

then:

$$\mathcal{I}_n(\theta) = \mathbb{E}\left\{\frac{\partial}{\partial\theta}\ln f(\mathbf{y}|\theta)\right\}^2 = -\mathbb{E}\left\{\frac{\partial^2}{\partial\theta^2}\ln f(\mathbf{y}|\theta)\right\}. \quad (3.10)$$

■

Remark 3.2. For exponential family densities (2.1), the sample Fisher information coincides with the negative expected Hessian of the sample log likelihood (3.10). $\blacklozenge$

Theorem 3.4 (Attainment). Suppose $\mathbf{Y} = (Y_1, \dots, Y_n)$ is an IID random sample, and that the CRLB condition (3.7) holds. An estimator T attains the CRLB for estimating $\tau(\theta) \iff$ the sample score is expressible as:

$$\mathcal{U}_n(\theta) = \frac{\partial}{\partial\theta}\ln f(\mathbf{y}|\theta) = a(\theta)\{T(\mathbf{y}) - \tau(\theta)\},$$

for some function $a(\theta)$ not depending on $\mathbf{y}$. $\square$

Proof. Proof of the CRLB made use of the Cauchy-Schwarz inequality (1.1). Letting $X = T$ and $Y = \mathcal{U}_n(\theta)$:

$$\mathbb{C}^2\{T, \mathcal{U}_n(\theta)\} \leq \mathbb{V}(T) \cdot \mathbb{V}\{\mathcal{U}_n(\theta)\}.$$

Equality is attained if and only if:

$$\mathcal{U}_n(\theta) = aT + b.$$

Further, since the sample score must have expectation zero:

$$0 = \mathbb{E}\{\mathcal{U}_n(\theta)\} = a\tau(\theta) + b \implies b = -a\tau(\theta).$$

■

3.4 Rao Blackwell

Theorem 3.5 (Rao Blackwell). Suppose $\tilde{\theta}$ is unbiased for θ , and that T is sufficient for θ . Define $\hat{\theta} = \mathbb{E}(\tilde{\theta}|T)$, then $\hat{\theta}$ is also unbiased for θ and:

$$\mathbb{V}(\hat{\theta}) \leq \mathbb{V}(\tilde{\theta}).$$

That is, $\hat{\theta}$ is a uniformly better estimator than $\tilde{\theta}$. □

Proof. By the definition of sufficiency, the distribution of the data $\mathbf{Y}$ given T does not depend on θ . Since $\tilde{\theta} = \tilde{\theta}(\mathbf{y})$ is a function of $\mathbf{y}$ only, the expectation:

$$\hat{\theta} = \mathbb{E}\{\tilde{\theta}(\mathbf{Y})|T\},$$

is in fact a *statistic*.

By iterated expectation:

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}\{\mathbb{E}(\tilde{\theta}|T)\} = \mathbb{E}(\tilde{\theta}) = \theta.$$

By law of total variance:

$$\begin{aligned} \mathbb{V}(\tilde{\theta}) &= \mathbb{V}\{\mathbb{E}(\tilde{\theta}|T)\} + \mathbb{E}\{\mathbb{V}(\tilde{\theta}|T)\} \\ &= \mathbb{V}(\hat{\theta}) + \mathbb{E}\{\mathbb{V}(\tilde{\theta}|T)\} \geq \mathbb{V}(\hat{\theta}). \end{aligned}$$

■

Example 3.1. Suppose $Y_i \stackrel{\text{IID}}{\sim} f(\mathbf{y})$, continuous but not necessarily parametric. An individual observation Y_i is unbiased for the mean $\mathbb{E}(Y_i) = \mu$. The sample order statistics $(Y_{(1)}, \dots, Y_{(n)})$ are always sufficient.

Applying the Rao Blackwell theorem to $\tilde{\theta} = Y_i$ and $T = (Y_{(1)}, \dots, Y_{(n)})$:

$$\hat{\theta} = \mathbb{E}(Y_i|Y_{(1)}, \dots, Y_{(n)}) \stackrel{*}{=} \frac{1}{n} \sum_{i=1}^n Y_{(i)} = \bar{Y}.$$

Equality $\stackrel{*}{=}$ holds since the distribution of Y_i given all order statistics is discrete uniform on $Y_{(1)}, \dots, Y_{(n)}$. ♠

Theorem 3.6 (Lehmann Scheffe). If T is a *complete sufficient statistic*, then $h(T)$ is the UMVUE of its expectation, provided $\mathbb{V}\{h(T)\} < \infty$. □

Proof. Suppose $\tilde{\theta}_1$ and $\tilde{\theta}_2$ are two unbiased estimators of θ . Since T is sufficient, by the Rao Blackwell theorem $\hat{\theta}_1 = \mathbb{E}(\tilde{\theta}_1|T)$ and $\hat{\theta}_2 = \mathbb{E}(\tilde{\theta}_2|T)$ are two unbiased estimators of θ with variance no greater than the original estimators. Define:

$$g(T) = \hat{\theta}_1 - \hat{\theta}_2 = \mathbb{E}(\tilde{\theta}_1|T) - \mathbb{E}(\tilde{\theta}_2|T).$$

Since $\mathbb{E}\{g(T)\} = 0$ and T is complete, conclude that $\hat{\theta}_1 = \hat{\theta}_2$. Thus, given any initially unbiased estimator $\tilde{\theta}$, the unique estimator $\hat{\theta} = \mathbb{E}(\tilde{\theta}|T)$ is also unbiased and satisfies $\mathbb{V}(\hat{\theta}) \leq \mathbb{V}(\tilde{\theta})$. If $\mathbb{V}(\hat{\theta}) < \infty$, then $\hat{\theta}$ is the UMVUE (if not, there may be multiple best estimators of θ).

Now, consider estimating $\mathbb{E}\{h(T)\}$ by $h(T)$. Observe that $h(T)$ is unbiased and that $h(T) = \mathbb{E}\{h(T)|T\}$. Provided $\mathbb{V}\{h(T)\} < \infty$, $h(T)$ is the UMVUE of its expectation. ■

3.5 Exercises

- i. Find the log likelihood, score, and Fisher information for IID random samples from the following exponential family distributions:
 - (a) Binomial.
 - (b) Poisson.
 - (c) Normal.
 - (d) Gamma.
- ii. Prove the bias-variance decomposition (3.6).
- iii. Prove that the sample information is the negative expected Hessian of the sample log likelihood (3.10).
- iv. Prove that the sample mean is the UMVUE for these distributions:
 - (a) Binomial.
 - (b) Poisson.

Hypothesis Testing

4.1 Likelihood Ratio

Definition 4.1. The likelihood ratio for testing $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is:

$$\lambda(\mathbf{Y}) = \frac{\sup_{\theta \in \Theta_0} L(\theta|\mathbf{Y})}{\sup_{\theta \in \Theta_1} L(\theta|\mathbf{Y})}. \quad (4.1)$$

■

Theorem 4.1. If $T = T(\mathbf{Y})$ is a sufficient statistic for θ , then the likelihood ratio statistic in (4.1) is expressible as:

$$\lambda(\mathbf{Y}) = \frac{\sup_{\theta \in \Theta_0} L\{\theta|T(\mathbf{Y})\}}{\sup_{\theta \in \Theta_1} L\{\theta|T(\mathbf{Y})\}}. \quad (4.2)$$

□

Remark 4.1. (4.2) indicates that the likelihood ratio statistic for θ should depend on the sample $\mathbf{Y}$ only through a sufficient statistic for θ . ◆

Example 4.1. Suppose $Y_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, and that interest lies in making inferences about μ , with σ^2 regarded as a nuisance parameter. In particular, consider evaluating the $H_0 : \mu = \mu_0$ versus $H_A : \mu \neq \mu_0$. The likelihood ratio statistic is:

$$\lambda(\mathbf{y}) = \frac{\sup_{\sigma^2 \in (0, \infty)} L(\mu_0, \sigma^2 | \mathbf{y})}{\sup_{\mu \in \mathbb{R}, \sigma^2 \in (0, \infty)} L(\mu, \sigma^2 | \mathbf{y})} = \frac{L(\mu_0, \tilde{\sigma}_0^2 | \mathbf{y})}{L(\hat{\mu}, \hat{\sigma}^2 | \mathbf{y})},$$

where $\tilde{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mu_0)^2$ and $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n y_i$, $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\mu})^2$. ♠

4.2 Power

Definition 4.2. Define the **rejection region** $\mathcal{R}$ as the subset of the sample space $\mathcal{Y}$ for which a hypothesis test ϕ rejects:

$$\mathcal{R} = \{\mathbf{y} \in \mathcal{Y} : \phi \text{ rejects}\}.$$

The **retention region** $\mathcal{A}$ is the subset of the sample space for which the hypothesis test fails to reject:

$$\mathcal{A} = \{\mathbf{y} \in \mathcal{Y} : \phi \text{ does not reject}\}.$$
■

Definition 4.3. The **power function** $\beta(\theta)$ is the probability that a sample falls in the rejection region as a function of the true parameter θ :

$$\beta(\theta) = \mathbb{P}_\theta(\mathbf{Y} \in \mathcal{R}).$$
■

Remark 4.2. The power function of the ideal test is equal to zero for $\forall \theta \in \Theta_0$, and equal to one for $\forall \theta \in \Theta_1$. ◆

Definition 4.4. A **type I error** is the probability of rejecting the null hypothesis when the null hypothesis is true:

$$\text{Type I Error} = \mathbb{P}_\theta(\mathbf{Y} \in \mathcal{R}) \text{ when } \theta \in \Theta_0.$$

A **type II error** is the probability of retaining the null hypothesis when the null hypothesis is false:

$$\text{Type II Error} = \mathbb{P}_\theta(\mathbf{Y} \in \mathcal{A}) \text{ when } \theta \in \Theta_1.$$
■

Definition 4.5. A test is described as **size** α if:

$$\sup_{\theta \in \Theta_0} \beta(\theta) = \alpha.$$

By contrast, a test is **level** α if:

$$\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha.$$

Every size α test is also level α . ■

Example 4.2. Suppose $Y_i \stackrel{\text{IID}}{\sim} N(\mu, \sigma^2)$, with σ^2 assumed known. The LRT of $H_0 : \mu \leq \mu_0$ against $H_A : \mu > \mu_0$ rejects if:

$$\frac{(\bar{Y} - \mu_0)}{\sigma/\sqrt{n}} > \zeta_{1-\alpha}.$$

The power of this test is:

$$\begin{aligned} \beta(\mu) &= \mathbb{P} \left\{ \frac{(\bar{Y} - \mu_0)}{\sigma/\sqrt{n}} > \zeta_{1-\alpha} \right\} = \mathbb{P} \left\{ \frac{(\bar{Y} - \mu + \mu - \mu_0)}{\sigma/\sqrt{n}} > \zeta_{1-\alpha} \right\} \\ &= \mathbb{P} \left\{ \frac{(\bar{Y} - \mu)}{\sigma/\sqrt{n}} > \zeta_{1-\alpha} - \frac{(\mu - \mu_0)}{\sigma/\sqrt{n}} \right\} = 1 - \Phi \left\{ \zeta_{1-\alpha} - \frac{(\mu - \mu_0)}{\sigma/\sqrt{n}} \right\}. \end{aligned}$$

♠

4.3 Neymann-Pearson

Definition 4.6. Consider a class $\mathcal{C}$ of tests for evaluating $H_0 : \theta \in \Theta_0$ against the alternative $H_A : \theta \in \Theta_A$. A test with power function $\beta(\theta)$ is **uniformly most powerful** if $\beta(\theta) \geq \tilde{\beta}(\theta)$ for every $\theta \in \Theta_A$ and every power function $\tilde{\beta}$ belonging to a test in $\mathcal{C}$. ■

Theorem 4.2 (Neymann-Pearson Lemma). Consider testing $H_0 : \theta = \theta_0$ against $H_A : \theta = \theta_1$. Suppose the rejection region takes the form:

$$\mathcal{R} = \{ \mathbf{y} \in \mathcal{Y} : f(\mathbf{y}|\theta_1) > k_\alpha f(\mathbf{y}|\theta_0) \}, \quad (4.3)$$

where k_α is chosen such that:

$$\mathbb{P}_{\theta_0}(\mathbf{Y} \in \mathcal{R}) = \alpha. \quad (4.4)$$

- i. Any test with rejection region (4.3) that satisfies (4.4) is a UMP level- α test.
- ii. If the preceding test exists, then every UMP level- α is size- α and has a rejection region that agrees with (4.3) a.e.

□

Remark 4.3. Note that k_α is chosen such that the probability $\mathbf{Y}$ falls in the rejection region $\mathcal{R}$ is α under the null hypothesis $H_0 : \theta = \theta_0$. ♦

Proof. (i.) Define the **test function**:

$$\phi(\mathbf{y}) = \mathbb{I}(\mathbf{y} \in \mathcal{R}),$$

where $\mathcal{R}$ is defined in (4.3) and satisfies (4.4). Let $\tilde{\phi}$ denote the test function of any other level- α test; let β and $\tilde{\beta}$ denote the corresponding power functions. The power function β is related to the test function via:

$$\beta(\theta) = \mathbb{P}_\theta(\mathbf{Y} \in \mathcal{R}) = \mathbb{E}_\theta\{\mathbb{I}(\mathbf{Y} \in \mathcal{R})\} = \int \phi(\mathbf{y}) dF(\mathbf{y}; \theta).$$

The function $0 \leq g(\mathbf{y}) = \{\phi(\mathbf{y}) - \tilde{\phi}(\mathbf{y})\}\{f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0)\}$ is non-negative since $\tilde{\phi} \in \{0, 1\}$, $\phi(\mathbf{y}) = 1$ if $f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0) > 0$ and $\phi(\mathbf{y}) = 0$ if $f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0) < 0$. Now:

$$\begin{aligned} 0 &\leq \int g(\mathbf{y}) d\mathbf{y} = \int \{\phi(\mathbf{y}) - \tilde{\phi}(\mathbf{y})\}\{f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0)\} d\mathbf{y} \\ &= \beta(\theta_1) - \tilde{\beta}(\theta_1) - k_\alpha\{\beta(\theta_0) - \tilde{\beta}(\theta_0)\}. \end{aligned}$$

Since ϕ is size- α and $\tilde{\phi}$ is level- α , $\beta(\theta_0) - \tilde{\beta}(\theta_0) \geq 0$, hence:

$$0 \leq \beta(\theta_1) - \tilde{\beta}(\theta_1) - k_\alpha\{\beta(\theta_0) - \tilde{\beta}(\theta_0)\} \leq \beta(\theta_1) - \tilde{\beta}(\theta_1).$$

(ii.) Suppose ϕ is defined as previously, and $\tilde{\phi}$ is another UMP level- α test. Since ϕ and $\tilde{\phi}$ are both UMP, $\beta(\theta_1) - \tilde{\beta}(\theta_1) = 0$. From the above, conclude that:

$$0 = \beta(\theta_0) - \tilde{\beta}(\theta_0) = \alpha - \tilde{\beta}(\theta_0).$$

That is, $\tilde{\phi}$ is also size- α . Consequently,

$$\int g(\mathbf{y}) d\mathbf{y} = \int \{\phi(\mathbf{y}) - \tilde{\phi}(\mathbf{y})\}\{f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0)\} d\mathbf{y} = 0.$$

Since $f(\mathbf{x}|\theta_1) - k_\alpha f(\mathbf{x}|\theta_0) \neq 0$, the integral vanishes $\iff \phi(\mathbf{y}) = \tilde{\phi}(\mathbf{y})$ a.e. ■

Corollary 4.1. Consider again testing $H_0 : \theta = \theta_0$ against $H_A : \theta = \theta_1$. Suppose T is sufficient for θ , and $g(t|\theta)$ is the density of the sufficient statistic. A test based on T is UMP level- α if it satisfies:

$$\mathcal{R} = \{t \in \mathcal{T} : g(t|\theta_1) > k_\alpha g(t|\theta_0)\},$$

where k_α is chosen such that:

$$\mathbb{P}_{\theta_0}(T \in \mathcal{T}) = \alpha.$$

♣

4.4 Karlin-Rubin

Definition 4.7. A family of densities $g(t|\theta)$ for a univariate random variable T has a **monotone likelihood ratio** if for $\theta_2 > \theta_1$, the ratio:

$$\frac{g(t|\theta_2)}{g(t|\theta_1)}, \quad (4.5)$$

is **non-decreasing** as a function of t . ■

Proposition 4.1. Suppose T has a monotone likelihood ratio (4.5), then T is **stochastically non-decreasing** in θ . That is, for $\theta_2 > \theta_1$:

$$G(t|\theta_2) \leq G(t|\theta_1), \quad (4.6)$$

◆

Proof. Define $H(t) = G(t|\theta_2) - G(t|\theta_1)$. The derivative is:

$$\frac{d}{dt}H(t) = g(t|\theta_2) - g(t|\theta_1) = g(t|\theta_1) \left(\frac{g(t|\theta_2)}{g(t|\theta_1)} - 1 \right).$$

Since $g(t|\theta_1) > 0$ and $g(t|\theta_2)/g(t|\theta_1)$ is non-decreasing, the derivative of $H(t)$ can only change sign from negative to positive. Therefore, any interior critical point of $H(t)$ is a minimum, and the maximum of $H(t)$ must occur at the boundaries, $\{-\infty, \infty\}$. By the properties of distribution functions, $H(-\infty) = 0$ and $H(\infty) = 0$. Conclude that $H(t) \leq 0 \implies G(t|\theta_2) \leq G(t|\theta_1)$. ■

Corollary 4.2. If T has a monotone likelihood ratio, then for $\theta_2 > \theta_1$:

$$\mathbb{P}_{\theta_2}(T > t) \geq \mathbb{P}_{\theta_1}(T > t). \quad (4.7)$$

♣

Theorem 4.3. Consider testing $H_0 : \theta \leq \theta_0$ against $H_A : \theta > \theta_0$. Suppose T is sufficient for θ , and that $g(t|\theta)$ has a monotone likelihood ratio. Then, for any t_0 , the test with rejection region:

$$\mathcal{R} = \{t \in \mathcal{T} : t > t_0\},$$

is a UMP level- α test, where $\alpha = \mathbb{P}_{\theta_0}(T > t_0)$. □

Proof. Let $\beta(\theta) = \mathbb{P}_{\theta}(T > t_0)$ denote the power function. By (4.7), $\beta(\theta)$ is non-decreasing. Therefore:

$$\sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0) = \mathbb{P}_{\theta_0}(T > t_0),$$

demonstrating this is a level $\alpha \equiv \mathbb{P}_{\theta_0}(T > t_0)$ test.

Fix $\theta_1 > \theta_0$, and define:

$$k = \inf_{t \in \mathcal{U}} \frac{g(t|\theta_1)}{g(t|\theta_0)},$$

where $\mathcal{U}$ is the set:

$$\mathcal{U} = \{t \in \mathcal{T} : t > t_0 \text{ and either } g(t|\theta_1) > 0 \text{ or } g(t|\theta_0) > 0\}.$$

Now $t > t_0 \iff g(t|\theta_1) > kg(t|\theta_0)$. By the corollary to the Neymann-Pearson lemma, the test with rejection region:

$$\mathcal{R} = \{t \in \mathcal{T} : t > t_0\} = \{t \in \mathcal{T} : g(t|\theta_1) > kg(t|\theta_0)\},$$

is UMP for testing $H_0 : \theta = \theta_0$ against $H_A : \theta = \theta_1$. Since $\theta_1 > \theta_0$ was arbitrary, the test is UMP for $\forall \theta > \theta_0$. ■

Corollary 4.3. If T is sufficient for θ and $g(t|\theta)$ has a monotone likelihood ratio, then the rejection region of the UMP test of $H_0 : \theta \geq \theta_0$ against $H_A : \theta < \theta_0$ takes the form:

$$\mathcal{R} = \{t \in \mathcal{T} : t < t_0\},$$

where $\alpha = \mathbb{P}_{\theta_0}(T < t_0)$. ♣

4.5 p-Values

Definition 4.8. A **p-value** is a *statistic* $p(\mathbf{y}) \in [0, 1]$ such that p approaching zero provides increasing evidence against H_0 . A p-value is **valid** if:

$$\sup_{\theta \in \Theta_0} \mathbb{P}_{\theta}\{p(\mathbf{y}) \leq \alpha\} \leq \alpha.$$
■

4.6 Exercises

- i. Prove (4.2).
- ii. Prove the corollary to the Neymann-Pearson Lemma.
- iii. Verify (4.7).
- iv. Suppose $Y_i \sim N(\mu, \sigma^2)$ with σ^2 known.

- (a) Find the UMP, size α test of $H_0 : \mu \leq \mu_0$ against $H_A : \mu > \mu_0$.

- (b) Show that a UMP test of $H_0 : \mu = \mu_0$ v. $H_0 : \mu \neq \mu_0$ DNE.
- v. Suppose $Y_i \sim \text{Weibull}(\alpha, \lambda)$, with the shape α and rate λ parameters both unknown. Find the likelihood ratio test of $H_0 : \alpha = 1$ against $H_A : \alpha \neq 1$ in the presence of the nuisance parameter λ .

Confidence Intervals

5.1 Interval Estimators

Definition 5.1. An **interval estimator** of a scalar parameter θ is a pair of statistics $L(\mathbf{y})$ and $U(\mathbf{y})$, with $L(\mathbf{y}) \leq U(\mathbf{y})$, such that $L(\mathbf{y}) \leq \theta \leq U(\mathbf{y})$. ■

Definition 5.2. The **coverage probability** of an interval estimator is the probability the interval covers the true parameter θ :

$$\text{Coverage}(\theta) = \mathbb{P}_\theta\{(L \leq \theta) \cap (U \geq \theta)\}.$$

The **confidence coefficient** is the infimum of the coverage probability:

$$\gamma = \inf_{\theta \in \Theta} \mathbb{P}_\theta\{(L \leq \theta) \cap (U \geq \theta)\}.$$

Remark 5.1. In defining the coverage probability, the interval $[L, U]$, not the parameter, is random. In general, the coverage probability can depend on θ . When θ is unknown, we can only guarantee that the coverage probability is at least the confidence coefficient γ . ♦

5.2 Test Inversion

Example 5.1. Suppose $Y_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known. Consider testing $H_0 : \mu = \mu_0$ against $H_A : \mu \neq \mu_0$. The sample mean $\hat{\mu} = \bar{Y}$ is sufficient for μ . The rejection region of the UMP, unbiased, level- α test of $H_0 : \mu = \mu_0$ is:

$$\mathcal{R}(\mu_0) = \left(\mathbf{y} \in \mathcal{Y} : \frac{|\hat{\mu} - \mu_0|}{\sigma/\sqrt{n}} > z_{1-\alpha/2} \right).$$

Under H_0 , the probability that $\mathbf{Y}$ falls in the rejection region is:

$$\mathbb{P}_{\mu_0}\{\mathbf{Y} \in \mathcal{R}(\mu_0)\} = \mathbb{P}_{\mu_0}\left(\frac{|\hat{\mu} - \mu_0|}{\sigma/\sqrt{n}} > z_{1-\alpha/2}\right) = \alpha.$$

Equivalently, the probability that $\mathbf{Y}$ falls in the retention region is:

$$\mathbb{P}_{\mu_0}\{\mathbf{Y} \in \mathcal{A}(\mu_0)\} = \mathbb{P}_{\mu_0}\left(\frac{|\hat{\mu} - \mu_0|}{\sigma/\sqrt{n}} \leq z_{1-\alpha/2}\right) = 1 - \alpha.$$

Rearranging gives:

$$\mathbb{P}_{\mu_0}\left(\hat{\mu} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \hat{\mu} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Define:

$$L = \hat{\mu} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad U = \hat{\mu} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Now, $\mathbb{P}_{\mu_0}(L \leq \mu_0 \leq U) = 1 - \alpha$. Finally, observe that the last probability statement holds for every μ_0 . Thus (L, U) provides a $(1 - \alpha)$ confidence interval for μ . ♠

Discussion 5.1. Recall that the *rejection region* $\mathcal{R}$ is defined as the subset of the sample space $\mathcal{Y}$ for which a test ϕ rejects, and the *retention region* $\mathcal{A}$ is the subset of the sample space for which ϕ fails to reject. In general, the retention region $\mathcal{A}$ depend on the value θ_0 of the parameter under H_0 . In the previous example:

$$\mathcal{A}(\mu_0) = \left\{ \mathbf{y} \in \mathcal{Y} : \hat{\mu} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \hat{\mu} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \right\}$$

The corresponding **confidence set** is the subset of parameter space for which θ is a plausible value, given the data:

$$\mathcal{C}(\mathbf{y}) = \left\{ \mu \in \mathbb{R} : \hat{\mu} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \hat{\mu} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \right\}$$

The retention region and the confidence set are linked by the duality:

$$\mathbf{y} \in \mathcal{A}(\theta_0) \iff \theta_0 \in \mathcal{C}(\mathbf{y}).$$

♠

Theorem 5.1 (Duality). For each $\theta \in \Theta$, let $\mathcal{A}(\theta_0)$ denote the retention region of a level- α test of $H_0 : \theta = \theta_0$. Now, for each $\mathbf{y} \in \mathcal{Y}$, define the set:

$$\mathcal{C}(\mathbf{y}) = \{\theta \in \Theta : \mathbf{y} \in \mathcal{A}(\theta)\}.$$

Then $\mathcal{C}(\mathbf{y})$ is a $(1 - \alpha)$ confidence set for θ . Conversely, suppose $\mathcal{C}(\mathbf{y})$ is a $(1 - \alpha)$ confidence set for θ . For each $\theta \in \Theta$, define the set:

$$\mathcal{A}(\theta_0) = \{\mathbf{y} \in \mathcal{Y} : \theta_0 \in \mathcal{C}(\mathbf{y})\}.$$

Then $\mathcal{A}(\theta_0)$ is the retention region of a level- α test of $H_0 : \theta = \theta_0$. □

Example 5.2 (Inverting the LRT). Suppose $Y_i \stackrel{\text{iid}}{\sim} F_\theta$. The rejection region for a likelihood ratio test of $H_0 : \theta = \theta_0$ is:

$$\mathcal{R}(\theta_0) = [\mathbf{y} \in \mathcal{Y} : -2\{\ell_n(\theta_0) - \ell_n(\hat{\theta})\} > \chi_{1,1-\alpha}^2],$$

where $\hat{\theta}$ is the MLE of θ , and $\chi_{1,1-\alpha}^2$ is the critical value of the $\chi_1^2(0)$ distribution.

The retention region is:

$$\mathcal{A}(\theta_0) = [\mathbf{y} \in \mathcal{Y} : -2\{\ell_n(\theta_0) - \ell_n(\hat{\theta})\} \leq \chi_{1,1-\alpha}^2].$$

Viewing the sample as fixed and the retention region as a function of the parameter gives the corresponding confidence set:

$$\mathcal{C}(\mathbf{y}) = [\theta \in \Theta : -2\{\ell_n(\theta_0) - \ell_n(\hat{\theta})\} \leq \chi_{1,1-\alpha}^2].$$



Example 5.3 (P-Inversion). Suppose $Y_i \stackrel{\text{iid}}{\sim} F_\theta$. Let $P(\mathbf{y}; \theta_0)$ denote a p-value assessing $H_0 : \theta = \theta_0$ based on $\mathbf{y}$. For a level α test, the rejection region is:

$$\mathcal{R}(\theta_0) = \{\mathbf{y} \in \mathcal{Y} : P(\mathbf{y}; \theta_0) \leq \alpha\}.$$

The retention region is:

$$\mathcal{A}(\theta_0) = \{\mathbf{y} \in \mathcal{Y} : P(\mathbf{y}; \theta_0) > \alpha\}.$$

Viewing the sample as fixed and θ as variable, the confidence set is:

$$\mathcal{C}(\mathbf{y}) = \{\theta \in \Theta : P(\mathbf{y}; \theta) > \alpha\}.$$



Example 5.4 (Clopper-Pearson Interval). Suppose $Y_i \stackrel{\text{iid}}{\sim} \text{Bern}(\pi)$. The total number of successes $T = \sum_{i=1}^n Y_i$ is sufficient for π . Define the function:

$$u(\theta) = \mathbb{P}\{\text{Binom}(n, \theta) \leq t_{\text{obs}}\}.$$

$u(\theta)$ is the p-value for testing $H_0 : \pi \geq \theta$ against $H_A : \pi < \theta$, and $u(\theta)$ is a *decreasing* function of θ . An upper confidence bound is given by:

$$U = \sup \left\{ \theta \in (0, 1) : u(\theta) > \frac{\alpha}{2} \right\}.$$

U is the largest value of θ for which $H_0 : \pi \geq \theta$ fails to reject at level $(\alpha/2)$.

Reciprocally, define the function:

$$l(\theta) = \mathbb{P}\{\text{Binom}(n, \theta) \geq t_{\text{obs}}\}.$$

$l(\theta)$ is the p-value for testing $H_0 : \pi \leq \theta$ against $H_A : \pi > \theta$, and $l(\theta)$ is an *increasing* function of θ . A lower confidence bound is given by:

$$L = \inf \left\{ \theta \in (0, 1) : l(\theta) > \frac{\alpha}{2} \right\}.$$

L is the smallest value of θ for which $H_0 : \pi \leq \theta$ fails to reject at level $(\alpha/2)$.



5.3 Pivots

Definition 5.3. A **pivot** is a function $Q(\mathbf{Y}, \theta)$ of the data $\mathbf{Y}$ and parameter θ whose distribution no longer depend on θ . ■

Example 5.5. Suppose $Y_i \stackrel{\text{iid}}{\sim} U(0, \theta)$. A sufficient statistic for θ is the sample maximum $Y_{(n)} = \max(Y_1, \dots, Y_n)$. Recall that, $Y_i \stackrel{d}{=} \theta X_i$, $Y_{(n)} \stackrel{d}{=} \theta X_{(n)}$, and $X_{(n)} \sim \text{Beta}(n, 1)$. The quantity $X_{(n)} = \theta^{-1} Y_{(n)}$ is pivotal for θ . To construct a confidence interval, we seek constants a and b such that:

$$\mathbb{P}(a \leq \theta^{-1} Y_{(n)} \leq b) = \mathbb{P}(a \leq X_{(n)} \leq b) = \int_a^b n t^{n-1} dt = t^n \Big|_a^b = b^n - a^n \stackrel{\text{Set}}{=} 1 - \alpha.$$

Having obtained $a < b$ numerically, a confidence interval for θ is given by:

$$\mathbb{P} \left(\frac{Y_{(n)}}{b} \leq \theta \leq \frac{Y_{(n)}}{a} \right) = 1 - \alpha.$$



Example 5.6. Suppose $Y_i \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$. A sufficient statistic for λ is the sum total $S = \sum_{i=1}^n Y_i$. Recall that, $Y_i \stackrel{d}{=} \lambda^{-1} X_i$, $S = \lambda^{-1} T$, where $T = \sum_{i=1}^n X_i$, and that $T \sim \text{Gamma}(n, 1)$. The quantity $T = \lambda S$ is pivotal for λ . To construct a confidence interval, we seek constants a and b such that:

$$\mathbb{P}(a \leq \lambda S \leq b) = \mathbb{P}(a \leq T \leq b) = \frac{1}{\Gamma(n)} \int_a^b t^{n-1} e^{-t} dt \stackrel{\text{Set}}{=} 1 - \alpha.$$

Having obtained $a < b$ numerically, a confidence interval for λ is given by:

$$\mathbb{P} \left(\frac{a}{S} \leq \lambda \leq \frac{b}{S} \right) = 1 - \alpha.$$



5.4 Exercises

- i. Prove the duality of confidence sets and hypothesis tests.
- ii. Construct a confidence set for the rate λ of an exponential distribution by inverting the likelihood ratio test.
- iii. Suppose $Y_i \stackrel{\text{IID}}{\sim} \text{Poi}(\lambda)$. Construct a Clopper-Pearson type confidence interval for λ .
- iv. Suppose $Y_i \stackrel{\text{IID}}{\sim} \text{Bern}(\pi)$.
 - (a) Find the variance stabilizing transformation $g(\cdot)$ of Y_i .
 - (b) Use the variance stabilized random variable $Z_i = g(Y_i)$ to construct an asymptotic confidence interval for π .